

SALTED: A symmetry-adapted machine-learning program for predicting electron-densities in molecules and materials

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Summary

SALTED provides an open-source Python package for machine learning the quantum-mechanical electron density, $n(\mathbf{r})$, in molecular and condensed-phase systems based on input atomic coordinates and species (Grisafi et al., 2019; Grisafi, Lewis, et al., 2023; Lewis et al., 2021). The program adopts a linear atom-centered decomposition of the electron density, which makes it highly transferable across diverse atomistic configurations sharing similar chemical environments. Because of this representation choice, SALTED is naturally interfaced with state-of-the-art electronic-structure programs based on atomic orbitals, namely CP2K (Iannuzzi et al., 2026), FHI-aims (Abbott et al., 2026), and PySCF (Sun et al., 2020), from which reference electron-density data can be generated and used to train a model. The learning algorithm is based on a symmetry-adapted extension of Gaussian process regression (Grisafi et al., 2018), making SALTED especially efficient in small-data regimes. Thanks to the implementation of vector-field kernel functions (Rossi et al., 2025), SALTED can also learn the first-order response of the electron density to applied electric fields, $\partial n(\mathbf{r})/\partial \mathbf{E}$. The application of SALTED within computational workflows has already shown its utility in a wide variety of contexts, including the calculation of polarization vectors (Grisafi, Bussy, et al., 2023) and polarizability tensors (Rossi et al., 2025), the accurate evaluation of Coulomb forces in QM/MM molecular-dynamics simulations (Grisafi & Salanne, 2024), and electronic-structure studies of large-scale 2D materials (Lou et al., 2026).

Statement of need

Electronic structure methods can be predictive for the simulation of the properties of a wide variety of materials. Among these methods, density-functional theory (DFT) has, perhaps, become the most successful and popular, having solved outstanding problems in areas as diverse as biochemistry, catalysis, nanotechnology and quantum materials (Burke, 2012; Jones, 2015). The popularity of DFT is based on its ability to provide sufficient accuracy for many practical applications at a modest cost. However, the cost of these calculations can only be considered modest in comparison to other, more advanced, electronic structure methods. For the current landscape of data-driven material science and molecular discovery, allied to the

quest of achieving first-principles accuracy for larger systems and longer time scales, even the cost of DFT calculations becomes prohibitively large.

A machine-learning method that can predict the most fundamental quantity of DFT, namely the real-space electronic density, therefore holds immense potential to deliver a single model that can be used to calculate a multitude of downstream material properties at a small fraction of the cost of DFT, while maintaining very similar accuracy. In this paper, we describe the SALTED software package, which allows electronic density and electronic-density response predictions over an atomic basis consistent with the underlying electronic-structure architectures.

The availability of this open-source software package provides the atomistic-simulation community with a practical and extensible framework for constructing machine-learning models of the electron density and its linear response from first-principles reference data. SALTED is intended for researchers in computational chemistry, condensed-matter physics, materials science, and molecular simulation who wish to accelerate electronic-structure calculations while retaining direct access to physically meaningful electronic observables. By interfacing with widely used electronic-structure packages, SALTED can be embedded into established computational workflows. The package is designed both for users seeking efficient predictions of electron densities and derived properties, and for method developers interested in extending the methodology or in interfacing density predictions with downstream electronic-structure and multiscale simulation tools. SALTED facilitates reproducible research and supports the development of machine-learning models for electronic structure by providing an open, documented, and reusable implementation of electron-density prediction.

State of the field

Two main directions have been independently followed to construct ML models of the electron density that either directly sample $n(\mathbf{r})$ on a real-space 3D grid, or expand $n(\mathbf{r})$ over a linear atomic basis. Examples of grid-based ML models with published open-source packages are: DeepDFT, based on equivariant graph neural networks (Jørgensen & Bhowmik, 2022); a linear-regression framework based on Jacobi-Legendre many-body descriptors (Focassio et al., 2023); Charge3net, based on a high-order equivariant neural network (Koker et al., 2024); and an extension of FIREANN, where an efficient sampling strategy is devised to drastically reduce the number of grid-point evaluations and to further treat the response to applied fields (Feng et al., 2025). Albeit not targeting directly the electronic density, the Materials Learning Algorithm (MALA) package learns the local density of states on a grid, and reconstructs $n(\mathbf{r})$ from this quantity (Cangi et al., 2025).

When compared with grid-based approaches, ML models that represent the density on an atomic basis carry the advantage of greatly reducing the amount of data at the price of an acceptably small error due to the finite expressiveness of the basis functions. The SALTED package is the result of one of the earliest ML approaches developed to learn the electron density via a set of atomic coefficients (Grisafi et al., 2019). Specifically, $n(\mathbf{r})$ is expanded over a linear basis made of radial functions R_n^λ and spherical harmonics Y_μ^λ centered around each atom of the system:

$$n(\mathbf{r}) = \sum_{in\lambda\mu} c_{in\lambda\mu} \sum_{\mathbf{u}} R_n^\lambda(|\mathbf{r} - \mathbf{r}_i - \mathbf{u}|) Y_\mu^\lambda(\widehat{\mathbf{r} - \mathbf{r}_i - \mathbf{u}})$$

where i are the atomic indexes, $\mathbf{u}$ are the cell translation vector when considering periodic systems (Lewis et al., 2021), and $c_{in\lambda\mu}$ are the density expansion coefficients. The reference data for $c_{in\lambda\mu}$ can directly be obtained from resolution of the identity (RI) methods implemented in electronic-structure codes that are based on atomic orbitals, making SALTED easy to interface with codes using such basis sets and methods.

87 As a key difference from the models previously mentioned, SALTED is based on a symmetry-
88 adapted extension of Gaussian process regression (GPR). Spherical-tensor kernels (Grisafi
89 et al., 2018), $\mathbf{K}^\lambda$, rather than equivariant neural-networks, are computed to satisfy exact
90 $\mathcal{O}(3)$ symmetries (Grisafi et al., 2019), starting from atom-density features obtained using the
91 featomic library (Bigi et al., 2026). While the adoption of a GPR method limits the scalability
92 of SALTED over strongly heterogeneous datasets (Grisafi, Lewis, et al., 2023), the constrained
93 function space defined by the kernels, together with the physically interpretable nature of
94 the underlying atomistic features, helps avoid overfitting—thereby facilitating transferable
95 extrapolations of the electron density across different system sizes. Moreover, when compared
96 with data-hungry neural-network architectures, the data-efficiency of GPR-based approaches
97 becomes a great advantage when training the model on high-level electronic-structure methods,
98 for which only a small number of calculations ($\sim 10^{2-3}$) can be performed.

99 In addition, SALTED presents a couple of distinctive features. First, a suitable linear combination
100 of the kernels $\mathbf{K}^\lambda$ allows SALTED to learn the first-order response of the density to uniform
101 applied electric fields, $\partial n(\mathbf{r})/\partial E_k$, thereby ensuring the correct transformations of a 3D vector
102 field (Rossi et al., 2025). Second, the use of the featomic library enables the inclusion of
103 long-distance equivariants (LODE) features (Grisafi & Ceriotti, 2019), which are essential
104 for learning the nonlocal redistribution of $n(\mathbf{r})$ in highly-polarizable systems such as metallic
105 frameworks (Grisafi, Bussy, et al., 2023; Rossi et al., 2025).

106 We note that there are other ML packages that share the same philosophy as SALTED in
107 representing the electron density similarly to the equation above, albeit with different features
108 and capabilities: equivariant neural-network architectures based on the E3NN model (Rackers
109 et al., 2023); scdp, which augments the atom-centered basis with additional “virtual” orbitals
110 placed at non-atomic sites (e.g., bond midpoints) to improve expressivity, and regresses the
111 expansion coefficients with a high-capacity equivariant neural network (Fu et al., 2024);
112 and lastly, an equivariant neural-network architecture that predicts atom-centered density
113 coefficients as an intermediate representation for learning energies and forces (Bogojeski et al.,
114 2026).

115 Software design

116 SALTED is organized as a workflow in three stages: dataset preparation, model training, and
117 prediction (Figure 1). Each stage is configured through a dedicated section of a single `inp.yaml`
118 file, and each step within a stage is exposed as an independently invocable command, `python`
119 `-m salted.<step>`. GitHub Continuous Integration (CI) is enabled for testing the package:
120 unit tests cover the utility modules, and pipeline tests run the full workflow through every
121 electronic-structure interface.

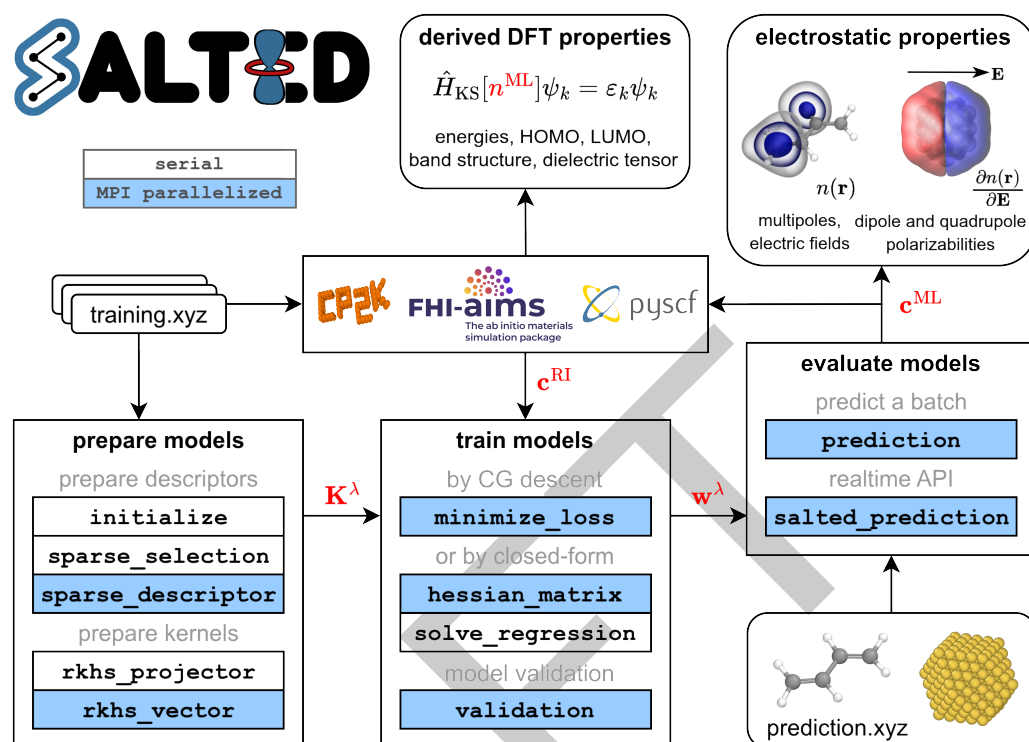


Figure 1: SALTED workflow. Reference density or density-response RI coefficients c^{RI} of the training dataset are generated by CP2K, FHI-aims, or PySCF. Model preparation builds λ -descriptors and assembles kernels K^λ of the training structures, with optional sparsification. Model training obtains regression weights w^λ using c^{RI} and K^λ . The trained models can be quickly validated to determine their accuracy, before being used to evaluate unseen structures. Predicted RI coefficients c^{ML} are used either directly, to evaluate electrostatic properties such as multipoles, electric fields, and polarizabilities, or as an input density from which the Kohn–Sham Hamiltonian is built and further obtain DFT properties after one diagonalization step. Shaded boxes mark stages that are parallelized by MPI.

This step-wise organization allows a run to be check-pointed, restarted, and distributed across separate HPC allocations, with the intermediate quantities of each step available on disk. Such control is necessary in practice: training a capable model over a large chemical space or on condensed-phase systems usually requires many HPC nodes, and the most efficient parallelization strategy differs from step to step. Users can resume an interrupted run, reuse descriptors across hyperparameter choices, and inspect intermediate quantities to improve models. We accept a less immediate interactive experience that a single-call library API would provide in exchange for workflows that remain manageable by the user.

SALTED integrates with a variety of electronic-structure codes based on atomic orbitals, CP2K (Iannuzzi et al., 2026), FHI-aims (Abbott et al., 2026), and PySCF (Sun et al., 2020), thus supporting a large fraction of community-developed electronic-structure software. Density data from CP2K are provided on uniform grids as *.cube files following pseudo-valence representations of $n(\mathbf{r})$; SALTED can then compute the required RI coefficients via a dedicated density_fitting module. FHI-aims is an all-electron electronic-structure program; in this case, density data for isolated and periodic systems are directly obtained as RI coefficients on the same footing. Finally, pseudo-valence PySCF density data for finite molecular systems can be obtained from SALTED by calling relevant PySCF functions.

SALTED unifies the training data obtained from these different codes within a single shared format. This requires extra engineering in the interfaces, but it lets research groups keep the electronic-structure code they already use in production, together with its established settings, and integrate SALTED into an existing workflow rather than switching electronic-

143 structure codes. CI tests cover all the interfaces. We note that both CP2K and PySCF work
 144 with Gaussian-type orbitals, enabling straightforward analytical calculations of density and
 145 density-response derived properties from the predicted SALTED coefficients, such as Hartree
 146 energies, electric-fields, dipoles and polarizabilities. While some of these properties are already
 147 computed and printed by SALTED upon density prediction, others could readily be implemented
 148 as dedicated post-processing modules.

149 The performance-critical steps, including equivariant descriptor contraction and the construction
 150 of Hessian matrix, are implemented as numba-compiled Python kernels (Lam et al., 2015). The
 151 numba kernels are carefully composed and benchmarked, and their portability does not come at
 152 the cost of performance. MPI parallelization is optional: parallelizable steps run serially by
 153 default and use mpi4py (Dalcin & Fang, 2021) automatically when launched under mpirun.
 154 OpenMP and Python thread pools would parallelize these steps within a single shared-memory
 155 node, whereas MPI covers the full range of hardware of the users, from personal computers and
 156 small workstations to multi-node HPC clusters. To improve portability, the supplied Dockerfile
 157 can be used to build a self-contained image with an MPI-enabled version of SALTED.

158 Research impact statement

159 The integration of SALTED predictions within computational workflows has already been shown
 160 useful in real-case scientific applications. For example, SALTED predictions of the charge-density
 161 response in metallic electrodes could be used to achieve a $\sim 10^3$ speedup in the calculation of
 162 DFT-level electrostatic forces entering the quantum-mechanics/molecular-mechanics simulation
 163 of model ionic capacitors (Grisafi, Lewis, et al., 2023). More recently, the use of the predicted
 164 density as a DFT input to perform a single Kohn-Sham Hamiltonian diagonalization has enabled
 165 the study of the electronic properties of twisted bilayer Moiré materials over extremely large
 166 supercells (Lou et al., 2026). Beyond electrostatic and DFT properties, SALTED enables the
 167 calculation of electron-density-based features at scale. A relevant example is the application
 168 to quantum-chemistry molecular interaction analysis (Fabrizio et al., 2020). Additionally, the
 169 RI density coefficients are directly applicable to crystal structure refinement from X-ray and
 170 electron diffraction data (Seifert et al., 2026), making the integration of SALTED predictions
 171 within these algorithms highly promising.

172 AI usage disclosure

173 We declare that no AI tool was used to make core code design decisions, nor were they used
 174 to make scientific decisions. Generative AI tools were employed to help with code refactoring
 175 and cleaning and with text polishing.

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